Multi-Objective Low-Carbon Scheduling Method for Data Centers Based on Ensemble Reinforcement Learning

Yifan Wang, *Graduate Student Member, IEEE*, Wei Sun, *Member, IEEE*, Pengyu Ren, *Graduate Student Member, IEEE*, and Gareth Harrison, *Senior Member, IEEE*

***Abstract*—The explosive growth of GPU data centers is driven by the surging demand for machine learning, artificial intelligence, and high-performance computing applications, with the pressure on the power grid intensifying. Meanwhile, as renewable energy becomes more integrated into the grid, the carbon intensity of grid electricity varies at different times. This variability presents an opportunity for data centers to align their workloads with periods of high renewable energy generation and low carbon intensity, thereby reducing their operational carbon emissions. This paper proposes a multi-objective scheduling strategy to minimize carbon emissions and maximize quality of service (QoS) by identifying the Pareto front. The proposed strategy employs an integrated reinforcement learning model to dynamically integrate action strategies for multiple objectives. Results from a case study based on a real-world, large-scale data center illustrate the method's effectiveness. This effectiveness arises from its dual-objective job scheduling strategy, which simultaneously reduces carbon emissions from electricity consumption and maintains a high quality of service in data centers. The method aims to promote the low-carbon operation of data centers, provide decision support, and facilitate coordination between data centers and power systems in real-world applications.**

***Index Terms*—Data center, job scheduling, demand side management, deep reinforcement learning, Pareto front.**

NOMENCLATURE

Abbreviations

CE	Carbon emission.
CI	Carbon intensity.
D3QN	Double dueling deep Q network.
QoS	Quality of service.
VC	Virtual cluster.

Set

$\mathcal{S}$	Set of state space, indexed by job.
A	Set of action space.

Symbols

α	Learning rate.
ϵ	Greedy parameter.
ϵ_{decay}	Decay factor.
γ	Discount factor.
$\Gamma(t)$	Scaled carbon intensity function.
τ	Interpolation parameter.
$a^{'}$	Action of the next step.
J_i^{at}	Arrival time of job i.
J_i^{gpu}	Number of GPUs required for job i.
J_i^{id}	Index for job i.
J_i^{L}	Job length.
J_i^{T}	Actual running time of the job i.
$N_t^{VC_i}$	Total number of virtual clusters for index i.
R_{CE}^{i}	Carbon emission reward for job i.
R_{QoS}^{i}	QoS reward for job i.
r_t	Reward at time step t.
$s^{'}$	State of the next step.
$T_{i,k}^{wait}$	Waiting time of the job i in virtual cluster k.
VC_j^{CI}	Scaled carbon intensity factor of the virtual cluster j.
VC_j^{eff}	Computational efficiency of the virtual cluster j.
VC_j^{id}	Index for the virtual cluster j.
VC_j^{num}	Available numbers of GPUs of the virtual cluster j.

I. INTRODUCTION

THE trend towards more complex computational jobs, especially training large language AI models such as ChatGPT by OpenAI, has led to explosive deployment of GPU resources in data centers [1].

However, the rapid expansion of data centers has significantly increased energy consumption and carbon emissions. In 2022, data centers accounted for approximately 1.2% of global electricity usage, which equates to 240–340 terawatt-hours (TWh), and were responsible for 0.6% of global carbon emissions [2]. The new AI servers in NVIDIA's data centers are projected to consume more power than Argentina and Sweden by 2027, at around 85.4 TWh per year, according

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The authors are with the School of Engineering, The University of Edinburgh, EH9 3FB Edinburgh, U.K. (e-mail: w.sun@ed.ac.uk).

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to calculations [3]. Meanwhile, facing the problem of heavy demand and carbon emissions, the US Government Accountability Office (GAO) has required the agencies to close 78 data centers in 2022 [4].

To meet global carbon reduction targets, the future development of data centers will increasingly focus on adopting low-carbon strategies [5]. Achieving this goal can be approached from two perspectives:

One is considering the data center's total energy demand as a whole, and adopting various methods to align the existing power consumption profile of data centers with the renewable energy supply closely, with respect to carbon reduction. The main feature of this approach is that it does not involve modifying the individual computational job schedule within the data center. The research in this direction has included adopting a new data center development framework that considers the features of environment-dependent components [6], integrating intelligent carbon management systems [7], and leveraging locally abundant renewable energy sources [8]. Since there is considerable existing research on this direction [6], [7], [8], [9], [10], this aspect will not be covered in this paper.

The other perspective is to focus on enhancing computational job flexibility within the data centers themselves, enabling a more proactive response to the electricity supply profile. Within a data center, there exists adjustable time between computational job arrival and execution, enabling flexible scheduling to a certain degree for purposes like cost reduction or environmental impact mitigation. This research direction presents notable challenges due to the high volume of computational jobs in large data centers that require swift handling [11]. As outlined in the literature review below, There are still significant research gaps in implementing a low-carbon job-based scheduling method to to address both service quality and carbon reduction.

With the development of machine learning (ML) and artificial intelligence, recent research has combined deep reinforcement learning (DRL) algorithms with job scheduling decision-making in data centers. Tripathy et al. [12] presented a dynamic job scheduling method for multi-processor grid environments using Directed Search Optimization (DSO), demonstrating superior performance over contemporary algorithms through simulation results. Fernández-Cerero et al. [13] introduced a machine learning regression model called Boost, which uses gradient boosting to predict scheduling times and enhance resource management in hyper-scale cloud-computing data centers. Sharma and Garg [14] proposed an artificial neural network-based (ANN) approach for energy-efficient job scheduling in cloud data centers, with a supervised neural network to optimize resource allocation, achieving substantial improvements over traditional algorithms in terms of makespan, energy consumption, and execution overhead.

Specifically, in DRL-based scheduling, Liang et al. [15] introduced A2cScheduler, an actor-critic architecture designed for multi-resource job dispatching in heterogeneous environments. Ding et al. [16] applied a Q-learning-based framework to reduce energy consumption in cloud data centers through dynamic job scheduling. Building on the idea of energy-aware policies, Liu et al. [17] proposed a low-carbon job scheduling scheme from an information–energy nexus perspective. Their DRL-based approach jointly considers delay, energy cost, and carbon intensity, addressing heterogeneous workloads with job dependencies.

As GPU workloads and large-scale AI applications proliferate, scheduling strategies must increasingly balance computational performance with environmental considerations. Zhao and Wu [18] applied multi-agent reinforcement learning (MARL) to improve inter-server coordination in GPU cluster scheduling. Zhang et al. [19] proposed a MARL framework for geo-distributed data centers that optimizes both service quality and carbon efficiency when handling AI-generated content (AIGC).

In a complementary direction, Riepin et al. [20] investigated spatiotemporal load shifting across distributed data centers to better align computational demand with clean energy supply, offering a system-level approach that coordinates workloads at a macro scale.

Building on these foundations, our work focuses on fine-grained carbon-aware job scheduling within a single GPU data center. We adopt a multi-objective DRL framework that leverages real carbon intensity signals and job traces to construct a Pareto front of scheduling strategies. This design enables flexible decision-making that reflects diverse operational pReferences and supports practical deployment in large-scale GPU clusters under realistic constraints.

Despite recent progress, most existing DRL-based scheduling approaches adopt scalarized single-objective reward formulations, where objectives like energy usage, job latency, and emissions are combined into a single cost or penalty function. While this simplifies policy learning, it limits the model's adaptability to changing operational priorities—for example, shifting from latency-sensitive job handling to carbon-aware deferral during low-emission periods.

Moreover, most methods rely on fixed reward weightings throughout training, without constructing a full Pareto frontier of scheduling options. This restricts flexibility and interpretability, especially in real-world GPU data centers that require nuanced trade-offs between service performance and environmental goals. There remains limited exploration of DRL frameworks that explicitly support multi-objective policy learning under realistic, carbon-sensitive workload conditions.

To provide a clearer comparison of recent work in this area, Table I summarizes representative scheduling strategies for data centers from the literature. The comparison spans multiple aspects, including whether carbon emissions are considered, the optimization objective type, the type of evaluation data, and whether online deployment is supported. As shown in the table, most DRL-based methods either optimize a single reward scalar or only consider QoS, and few support multi-objective scheduling using real-world data in an online setting.

To fill the research gaps, we propose a multi-objective low-carbon scheduling model for large-scale GPU data centers. We have introduced a new objective of carbon emission reduction with the QoS objective by using the carbon intensity of the electricity supply as an indicator. The carbon intensity of electricity supply is the carbon emissions per unit hour of grid electricity supply, which vary at different times in the

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TABLE I
DATA CENTER SCHEDULING STRATEGIES PROPOSED IN RECENT LITERATURE

Ref	Concerns		Features				
	QoS	Carbon emission	Method	Scheduling unit	Multi-Objective	Test data	Online status
Proposed Model	✓	✓	EDRL	Job	✓	Real	Online
[21]	✗	✗	MILP	Data center	✓	Generated	Offline
[22]	✓	✗	DRL	Job	Cost-only	Real	Online
[17]	✓	✓	DRL (Q-Learn)	Job	Cost-only	Real	Online
[16]	✓	✗	DRL (Q-Learn)	Job	Cost-only	Generated	Online
[23]	✓	✗	DRL	Job	Cost-only	Generated	Online
[24]	✓	✓	ML	Data center	Carbon-only	Real	Offline
[13]	✓	✗	ML	Job	Cost-only	Generated	Online
[14]	✓	✗	ANN	Data center	Cost-only	Generated	Online
[15]	✓	✗	DRL (A2C)	Job	Cost-only	Generated	Online
[18]	✓	✗	MARL + GNN	Job	Cost-only	Generated	Online
[19]	✓	✓	MARL	Job	Cost-only	Generated	Online

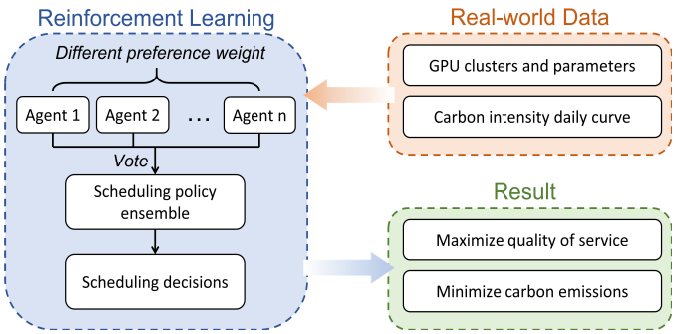


Fig. 1. Overview of the proposed GPU data center scheduling methodology.

day. This variability presents an opportunity for data centers to align their workloads with periods of high renewable energy generation and low carbon intensity, thereby reducing their operational carbon emissions.

To ensure real-time implementation, we employ a reinforcement learning-based algorithm due to its fast decision-making capabilities. As shown in Fig. 1, the diagram provides a system-level overview of the proposed scheduling framework. Multiple preference-weighted agents contribute to an ensemble scheduling policy that balances QoS and carbon emissions. In this workflow, GPU data center users will upload their jobs needed to be executed, which contain the required GPU numbers. Upon receiving this information, along with the real-time electricity carbon intensity value, the data center operator will use it as input to update the scheduling model. The job scheduling model has set the QoS and carbon emission as the double objectives for decision-making under resource constraints, which provides decision makers with a comprehensive view of the trade-offs involved in choosing a particular solution. The decision will determine the job assignment to specific virtual GPU clusters.

To the best of our knowledge, this paper is the first of its kind to propose a multi-objective Deep Reinforcement Learning algorithm based on Ensemble Learning (EL) to achieve real-time data center scheduling. The main contributions of this paper are as follows.

- We proposed an innovative data center scheduling algorithm to accomplish implementation scheduling under resource constraints. The objective of the model considers both the QoS of the jobs and the carbon intensity of the electricity supply at a certain time.
- We proposed an innovative way to combine Ensemble Learning with Deep Reinforcement Learning to search for the Pareto front in multi-objective decision-making based on rational weight allocation for different agents.
- Based on real-world data, we made a data center scheduling simulation case study. The results indicate that our trained model is capable of real-time data center scheduling and presents good convergence results.

The rest of this paper is structured as follows: Section II is the job scheduling framework, description, and modeling part. Section III is the EDRL training algorithm. Section IV is the case study based on real-world data. Section V is the conclusion.

II. JOB SCHEDULING FRAMEWORK DESCRIPTION AND MODELING

In this paper, our proposed framework mainly investigates the online job scheduling problem for low-carbon data center operations from both energy-saving and low-carbon perspectives. To address these two operational objectives simultaneously, on the one hand, the job scheduling policy is influenced by information-related factors, such as job arrival time and QoS. On the other hand, the job scheduling decision also depends on the carbon intensity of the electricity supply at a given time, which contributes to minimizing the data center's total carbon emissions.

A. Scheduling Model

As shown in Fig. 2, the principle framework of our EDRL algorithm can be described as follows:

The algorithm can described as the interaction between the agent and the environment. The agent makes decisions (action) based on the current state of the system, which includes various parameters and conditions relevant to the scheduling jobs in this model, such as job duration and virtual cluster operating conditions. The environment is basically a virtual place to react to the agent's decision, which results in rewards from the environment and feedback signals indicating the effectiveness of the decision taken. The cycle of state, action,

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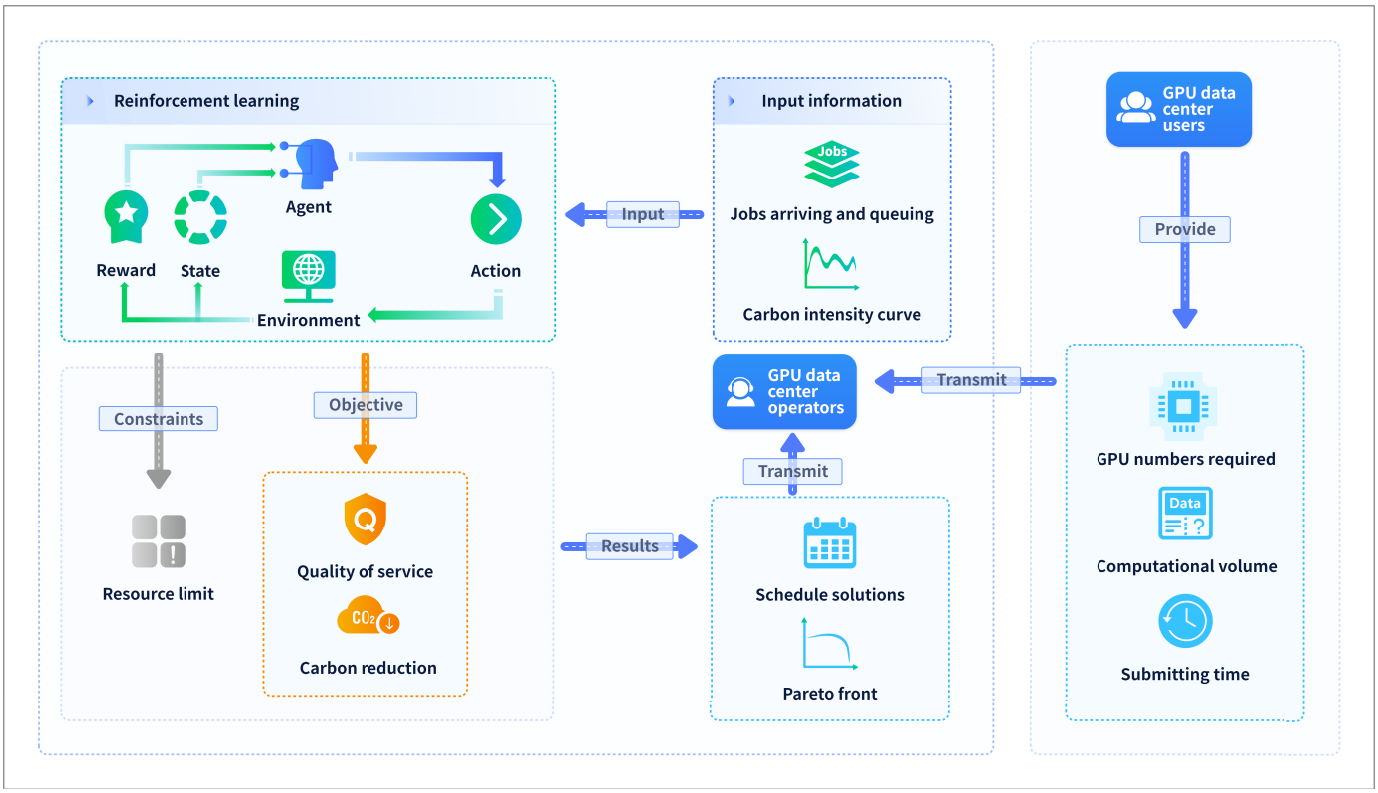


Fig. 2. Reinforcement learning-based scheduling loop and agent–environment interaction in the proposed model.

and reward forms the core learning mechanism of the DRL agent, continuously improving its decision-making process.

Initially, the GPU data center users upload their jobs, containing the information on the computational resources required for their jobs. The information that is passed to the data center side contains the number of GPUs required, the computational volume, and the submitting time.

The input information can be grouped into two primary parts: jobs arriving and queuing and the carbon intensity curve. The jobs arriving and queuing represent the computational jobs that need to be scheduled in the data center. The carbon intensity curve provides data on the variability of carbon emissions associated with the data center's energy consumption over time. The carbon intensity of the electricity supply could be provided by the energy supplier or the grid operator. This input information is crucial for the DRL agent to make informed scheduling decisions that balance between data center service performance and environmental impact.

Constraints play a significant role in shaping the scheduling strategy. The resource limit is a key constraint, indicating the maximum capacity and availability of computational resources, such as GPUs. These constraints ensure that the scheduling strategy operates within the equipment's feasible bounds, avoiding overcommitment of resources.

The objective of the scheduling strategy is twofold: optimizing the QoS and reducing carbon emissions. In the majority of cases, these two objectives tend to be contrary to each other: a system that prioritizes QoS typically follows a first-in-first-out (FIFO) strategy, which limits the flexibility to schedule the jobs effectively, resulting in smaller reductions in carbon emissions, and vice versa. Therefore, by considering the two objectives at the same time, the algorithm could find a trade-off between efficient and sustainable data center operations.

Finally, the output of the RL process is encapsulated in the results section, where the Pareto front illustrates the balance points between the different weights of the QoS and carbon reduction objectives. According to this, the GPU data center operators can make scheduling decisions through the obtained Pareto front,i.e., determine when to assign each job to which virtual cluster, taking into account the actual resource limit. Although the scheduling model operates in a non-preemptive manner, jobs may still experience deferred starts due to queuing conditions and resource availability. This scheduling flexibility allows the agent to align job execution with periods of lower carbon intensity when possible, without disrupting ongoing tasks.

B. Environment/Workloads

According to real data from the open-source data center log [25], we assume that the data center contains multiple virtual clusters, each of which includes several servers of the same type. In this work, we assume that all user-submitted jobs are independent and do not involve interdependencies or co-scheduling constraints. This modeling choice simplifies the scheduling environment and enables the policy to concentrate on carbon-aware dispatch decisions. Therefore, a job's properties can be summarized as tuple:

$$Job_i = (J_i^{id}, J_i^{at}, J_i^{gpu}, J_i^{L}) \in \mathbb{J} \tag{1}$$

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where the J_i^{id} is the job index generated sequentially when the job arrives at the data center, J_i^{at} is the job arrival time, J_i^{gpu} is the number of GPUs required set by the user, and J_i^L is the job's length.

The virtual clusters have attributes, including their name and the number of GPUs they have. In addition to simulating heterogeneous GPU characteristics and enabling more discriminative scheduling decisions, we introduce a virtual carbon intensity factor for each cluster, so it can be defined as:

$$VC_j = (VC_j^{id}, VC_j^{num}, VC_j^{CI}) \tag{2}$$

where the VC_j^{id} is the index for the virtual cluster, VC_j^{num} is the number of GPUs of the virtual cluster available for a selected day, and VC_j^{CI} is the scaled carbon intensity factor of GPUs in the cluster, which will be randomly generated between 1 and 1.2. This range ensures variability in the results when selecting different GPUs and enhances the uniqueness of the optimal solution.

This paper examines job scheduling to maximize total QoS and minimize CE over the course of a day. Before training, the jobs that started before the day are considered to have occurred and loaded into the environment. Subsequently, the optimal scheduling of jobs that arrive on that day will be analyzed, and the job scheduling policy will consider two objectives.

C. State Space

In this paper, the interaction between the environment and the scheduling model is based on the job index. When each job arrives, information such as job execution status and available computing resources collected by the real-time resources monitor will be shared with the scheduling model to help it make scheduling decisions. Specifically, the status information includes the attributes of the incoming work and the number of unused GPUs in each virtual cluster in the current data center. The state when the ith job arrives (i.e. $t = J_i^{at}$) can be expressed as follows:

$$\mathrm{S}_i = [J_i^{at}, J_i^{gpu}, J_i^L, N_t^{VC_1}, N_t^{VC_2}, \ldots, N_t^{VC_J}] \in \mathbb{S} \tag{3}$$

where $N_t^{VC_J}$ is the total number of idle GPUs of the VCs J in the data center at t, which is the time when ith job arrives. In a state S, the first three dimensions are the attributes of the ith job, and the following are the number of idle GPUs at time t for each cluster. Note that the job arrival time J_i^{at} and job length J_i^L are both measured in days. The GPU counts J_i^{gpu} and $N_t^{VC_J}$ are integer values without physical units.

D. Action Space

Action A_t represents the scheduling decision made by the scheduling model at the time step t, assigning the arrived job to a specific virtual cluster for execution. So, the action space can be expressed as:

$$\mathbb{A} = \{VC_1^{id}, VC_2^{id}, .., VC_J^{id}\} \tag{4}$$

Afterward, the selected virtual cluster processes the job by placing it in the queue, where it is executed in arrival order using either idle GPUs or GPUs that become available after completing previous jobs.

E. Reward Function

The scheduling model aims to minimize carbon emissions while maximizing QoS. A reward function is an estimate obtained by an agent performing certain actions on different states, with the results acting as timely feedback signals that interpret the quality of the current state or action. The reward function of QoS is the ratio of the job's length to the job's actual running time. The running time of the job can be calculated as:

$$J_i^T = \max_{k=1,\ldots,J_i^{gpu}} T_{i,k}^{wait} + J_i^L \tag{5}$$

For the QoS, we set the reward of QoS to:

$$R_i^{QoS} = \frac{J_i^L}{J_i^T} \tag{6}$$

It is obvious that this ratio is always less or equal to 1.

Then we scale it by adding 1 to it so that it becomes:

$$F(R_i^{QoS}) = 1 + R_i^{QoS} \tag{7}$$

The carbon emission of the job can be calculated as:

$$R_i^{CE} = VC_j^{CI} \sum_k^{J_i^{gpu}} \int_{et_k}^{st_k} \Gamma(t) \tag{8}$$

where J_i^T is the actual use time for the ith job, st_k and et_k are the start time and end time of the kth GPU branch required by the job, respectively. $\Gamma(t)$ is the carbon intensity factor curve on the selected day.

Since carbon intensity is affected by the job's running time and its magnitude varies greatly based on different jobs, we scale it through the log function. This transformation helps compress the unbounded scale of R_i^{CE}, which may vary from near-zero to very large values depending on job duration and carbon intensity. Compared with QoS rewards that are scaled into the [1, 2] range, this log-scaling ensures better numerical alignment and stability in optimization. The scaled reward function of carbon intensity can be expressed as:

$$G(R_i^{CE}) = \frac{1}{1 + \log(\mathrm{e} + R_i^{CE})} \tag{9}$$

Overall, the QoS reward $F(R_i^{QoS})$ ranges between 1 and 2, while the carbon emission reward $G(R_i^{CE})$ is derived through a logarithmic transformation to compress large-scale variations in raw carbon values, originally measured in grams of CO_2 per kWh (gCO_2/kWh). Both reward components are designed to be comparable in scale and free of physical units, ensuring numerical stability and effective joint optimization.

To reflect varying priorities during training, we combine these objectives into a single scalar reward using a multiplicative preference-based formulation. The scalarization approach and training process are detailed in Section III-B.

III. Proposed Multi-Objective Ensemble Deep Reinforcement Learning Algorithm

In multi-objective problems, multiple feedback signals need to be optimized simultaneously, and since there may be conflicts between the objectives, that is usually a set of

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trade-off strategies. This set of optimal (i.e., non-dominated, incomparable) strategies is called the Pareto front [26].

This paper proposes a multi-agent system ensemble reinforcement learning method whose agents prefer finding solutions in the Pareto front set. During the search process, each agent independently explores the Pareto front set based on different preference weights. The agents do not share experiences, replay buffers, or parameters, thereby ensuring diverse exploration across multiple Pareto-optimal solutions. Unlike traditional multi-agent reinforcement learning, our approach integrates multiple preference-weighted agents into a unified framework. These agents operate over the same environment and are co-trained within a shared decision-making loop, while each maintains distinct objective preferences.

A. Base Agent

The base agent adopts the Dueling Double Deep Q-network (D3QN) [27] algorithm, which is well-suited to our job scheduling scenario due to its high sample efficiency, stable training behavior, and compatibility with discrete action spaces. It combines two deep Q-network (DQN) variants: Double Deep Q-network [28] and Dueling Network [29] Architectures.

Q-learning is a popular reinforcement learning model used for making decisions in a discrete environment. It aims to determine the best action to take based on the current state. In this model, the agent keeps track of a value function, $Q(s, a)$, for each action-state pair and gets rewards from the action taken at state s. This function assists the agent in identifying the estimated Q-value of each action, allowing it to choose the action that will result in the highest reward in the long term. The Q-values are stored in a table called the Q table and are adjusted at each step. The Q-value is updated by:

$$\widehat{Q}(s, a) \leftarrow \widehat{Q}(s, a) + \alpha\delta \tag{10}$$

where δ is the temporal-difference error. Using different Q networks, the Double DQN algorithm separates action selection and evaluation in the target Q-value. The optimal target is as follows:

$$\delta = r + \gamma Q_{w^-}\left(s', \arg\max_{a' \in \mathbb{A}} Q_w(s'a')\right) \tag{11}$$

where r is reward, γ is discount factor, s' is the state of the next step, a' is the action of the next step, w is the weight of the current network, and $w-$ is the weight of the target network.

In the Dueling DQN algorithm, the network structure differs from the DQN, as it obtains intermediate features and adds two functions as action-value functions. By training V and A simultaneously, the loss will backpropagate through correct activation, so the overfitting problem during network training can be avoided. The action values can be obtained as follows:

$$Q_{\theta,\alpha,\beta}(s, a) = V_{\theta,\beta}(s, a) + A_{\theta,\alpha}(s, a) - \frac{\sum_{a'} A_{\theta,\alpha}(s, a')}{|\mathbb{A}|} \tag{12}$$

where s is current state, a is current action, θ is the of the shared network, and α and β represent the parameters of two branches.

B. Multi-Agent Ensemble

To approximate the Pareto front between carbon emission and QoS objectives, we employ an ensemble of reinforcement learning agents. Each agent is trained with a distinct fixed preference weighting that reflects a different trade-off between objectives. While each agent maintains its own network parameters, they are co-trained collaboratively during training through a unified decision process based on integrated action selection. This co-training strategy ensures the ensemble agent can explore diverse trade-off solutions effectively while maintaining coordination across agents. Compared to conventional DRL, which relies on a single agent learning a fixed reward preference, our ensemble framework simultaneously trains multiple agents with diverse reward weights. This design allows the model to approximate the full Pareto frontier rather than optimize toward a single weighted objective.

Unlike traditional multi-agent reinforcement learning, where agents interact with the environment independently and may require inter-agent communication, our agents operate collectively within a centralized framework. The ensemble behaves as a single agent from the environment's perspective, with all base policies sharing observations and decision context.

Instead of the single-objective problem, which returns a scalar-valued reward, the multi-objective problem returns a set of vector scalars corresponding to each m objective.

$$R(s, a, s') = (R_1(s, a, s'), \cdot;\cdot R_m(s, a, s')) \tag{13}$$

In this paper, we represent the different preferences of each base agent introduced in section A by giving each vector of rewards different magnitudes and multiplying them together:

$$P_{i,n} = \prod^{M} R_{i,m}^{\lambda_{m,n}} \tag{14}$$

where λ is the hyper-parameter used to trade off agents' preferences.

However, each agent will have different orders of magnitude of its Q-value based on different λ. On the other hand, considering the specific problem of this paper, there are large differences in the properties of each job (i.e., the length of the job and the number of GPUs required). To mitigate the impact caused by the differences from the environment (jobs), we scale it into a standard score before training by:

$$P_{i,n} \leftarrow Z(P_{i,n}) = \frac{P_{i,n} - \mu}{\sigma} \tag{15}$$

where μ and σ are the mean and standard deviation of P_n, respectively.

In addition, we scale the Q-value by dividing it by its modulus to avoid the impact caused by different Q-values trained by their own hyper-parameters:

$$Q_{i,n}(s, a) \leftarrow \frac{Q_{i,n}(s, a)}{|Q_{i,n}(s, a)|} \tag{16}$$

Overall, the pseudocode for the multi-agent ensemble of the reinforcement learning method is given in Algorithm 1.

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Algorithm 1 EDRL-based Job Scheduling Policy

Initialize current value network Q, target value network $\hat{Q}$, weight parameters of current network θ, weight parameters of target network $\hat{\theta}$, empty replay buffer D
Initialize probability of choosing to explore ϵ, environment state s_0
repeat
for each $Job_i \in \mathbb{J}$ **do**
$a_i \leftarrow \arg\max_a W \cdot Z(Q)^T$
Epsilon-Greedy action select action a_i with ϵ
Take action a_i, observe reward $r_i = (r_1, \ldots, r_m)$, new state s_i'
Add to buffer: $D \leftarrow D \cup (s_i, a_i, r_i, s_{i+1})$
for each T updates **do**
Sample a mini-batch B from D
for each learner $n = 1$ to N **do**
Calculate its reward by

$$Q = \prod_{i,m}^{M} r_{i,m}^{\lambda_{m,n}}$$

Compute the Q target y by

$$y = r + \gamma Q_{w^-}\left(s', \arg\max_{a' \in \mathbb{A}} Q_w(s'a')\right)$$

Update agent network parameter θ with gradient decent using

$$\nabla \frac{1}{|B|} \sum_{(s_i, a_i, r_i, s_{i+1}) \in B} \left(Q_{\theta,\alpha,\beta}(s,a) - y\right)^2$$

Soft update target network parameters with

$$\hat{\theta} \leftarrow \tau\hat{\theta} + (1-\tau)\theta$$

Update Q-value mean μ and standard deviation σ
end for
end for
$s \leftarrow s'$
end for
$\epsilon \leftarrow \epsilon \cdot \epsilon_{decay}$
until Max iteration

IV. Case Study

A. Basic Configuration

We used real-world data from Helios [30] to verify the effectiveness of our model. Helios is a private data center dedicated to developing DL models for research and production in SenseTime, which contains 8 independent virtual clusters and a total of more than 12,000 GPUs. To demonstrate the effectiveness of the proposed method, we selected 3 different typical days as cases, which have quite different numbers of jobs (varying from 254 in case 1 to 1541 in case 3) to be completed on the day. The basic configurations of the training data for three cases are shown in Table. II.

Note that once the study date is determined, we would begin by loading any jobs that started but were not completed before that day into the virtual clusters and then train the model using the data on that day. This electricity carbon intensity curve is sourced from the National Grid Electricity System Operator (ESO) for Great Britain [31]. The data for the corresponding study dates are plotted in Fig. 3. The electricity carbon intensity varies with the system's generation mix changes. The more renewable output at the time, indicating the percentage of renewable generation against the total generation at the time step as shown in the blue line in Fig. 3, the lower the carbon intensity. The demand also has an impact on it. During peak demand periods, increased output from thermal generation is needed to balance the demand, thus increasing the average carbon intensity of the system's electricity. The carbon intensity curve in Fig. 3 shows that the main daily peak in the carbon intensity curve occurs around 6 pm to 8 pm. This is because as the sun sets, solar generation decreases, and electricity demand begins to rise, causing the curve to rise sharply. The original carbon intensity data is reported at 30-minute intervals, and we adopt the same resolution to discretize the integration in Equation 8, approximating it as a summation over the job's runtime.

TABLE II
Basic Training Data Scale

	Date	Available GPU number	Job number	Job mean duration (hrs)
Case1	06-07	936	254	2.119
Case2	05-29	952	515	3.024
Case3	05-18	952	1102	1.382

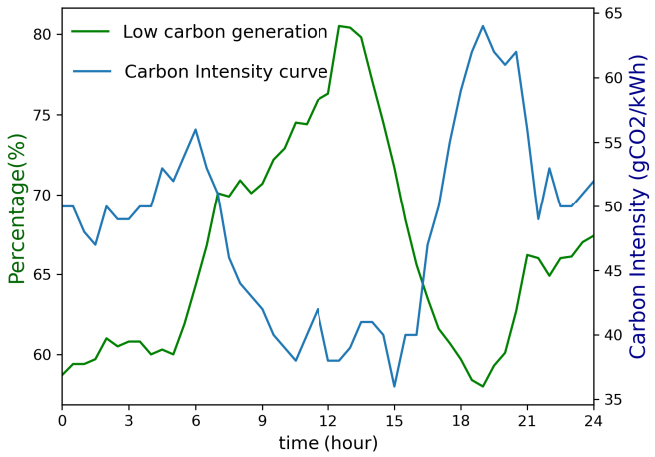


Fig. 3. Renewable energy percentage and carbon intensity curve of the electricity grid.

In this study, three base agents are instantiated, each with a fixed preference weighting (λ) that reflects a different trade-off between carbon emission reduction and QoS. The specific settings for λ are:

- (1.8 for carbon emission, 0.2 for QoS),
- (1.0 for carbon emission, 1.0 for QoS),
- (0.2 for carbon emission, 1.8 for QoS).

These configurations are selected to ensure broad coverage of possible trade-offs along the Pareto frontier. During training, the base agents are co-trained collaboratively, while during

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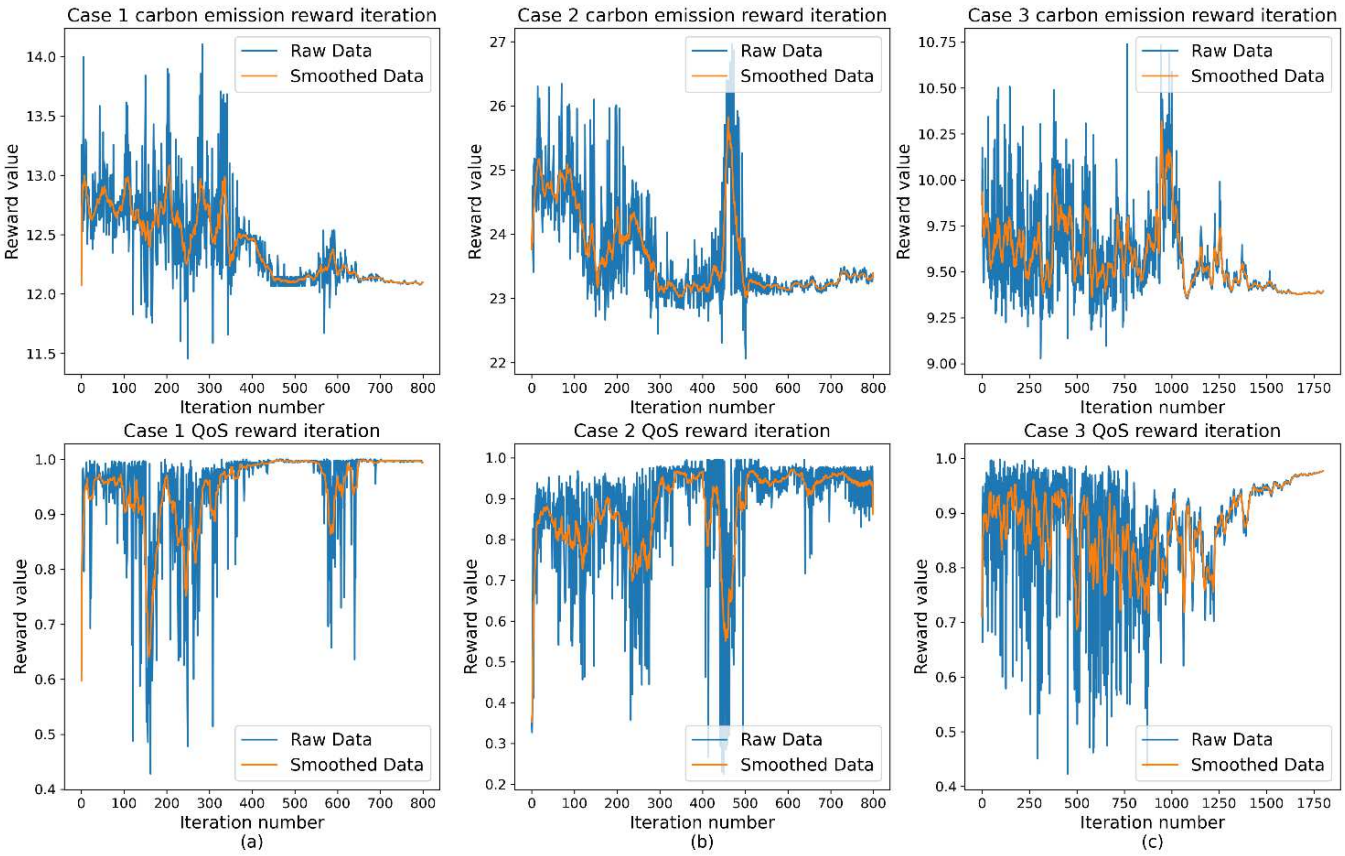


Fig. 4. EDRL carbon emission and QoS reward function iteration results.

inference, their outputs are integrated into a unified scheduling decision without requiring dynamic adjustments to λ.

Each base agent has an independent neural network, which consists of three fully connected hidden layers, each containing 128 neurons. Additionally, the value and advantage networks have two hidden layers with 64 and 32 neurons. The activation function used is the rectified linear activation function (ReLU). The optimizer is the Adam optimizer, and the learning rate α is 0.0001. The discount factor γ is set to 0.99, and the interpolation parameter τ is set to 0.01. The greedy parameter ϵ is decreased from 0.4 to 0.1, while the decay factor ϵ_{decay} is set to 0.995.

In this evaluation, we compare our scheduling results against real-world logs from the Helios data center, which reflect actual operator decisions under production conditions. This serves as a strong and practical baseline, as these logs inherently capture real workload dynamics and operational constraints.

Fig. 4 shows the convergence trends of the two reward functions for three different cases during the iteration process. To further illustrate the training dynamics and convergence behavior, we analyze the reward fluctuations and stabilization process in detail below. For the convenience of comparison, we took the average value of the total reward based on the number of jobs on the day of the case. As mentioned before, the goal of the model is to minimize the carbon emission reward while maximizing the QoS reward. Therefore, in the upper figure, the carbon intensity reward is expected to decrease to a lower value, while in the lower figure, a higher QoS reward is preferred.

In the left subfigure, the carbon intensity reward initially fluctuates between about 12 and 14. These fluctuations gradually decrease over time and stabilize around 12. Similarly, the QoS reward initially fluctuates widely between 0.4 and 1.0 but gradually stabilizes around 1.0 as the fluctuation decreases, indicating convergence.

In the middle subfigure, the carbon intensity reward initially fluctuates between about 22 and 27 and then stabilizes around 23. The QoS reward follows a similar pattern, initially fluctuating between 0.4 and 1.0 and gradually stabilizing around 1.0 as the fluctuation decreases.

B. Analysis of Optimization Process

As shown in Fig. 4, the convergence trends of the carbon intensity and QoS rewards across three case studies. In the right subgraph, the carbon intensity reward fluctuates between 9 and 11 and then stabilizes at about 9.4. Similarly, the QoS reward initially fluctuates greatly between 0.4 and 1.0 and then gradually stabilizes at about 1.0 as the fluctuation decreases.

In the initial stage, due to the high greediness rate, the agent tends to randomly select actions, resulting in large fluctuations in the algorithm and large fluctuations in rewards. However, with enough attempts, the algorithm may also obtain higher rewards. As the greediness rate decreases in the later stage,

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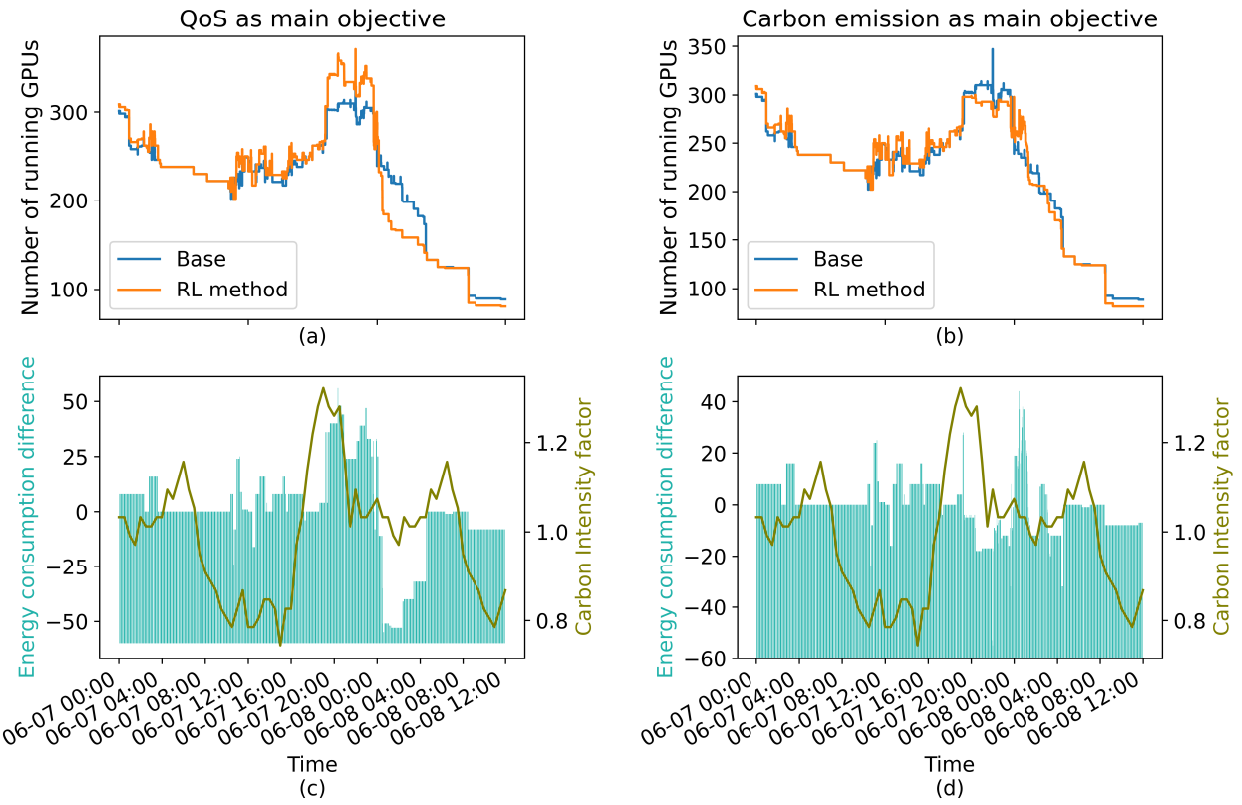


Fig. 5. Data center scheduling results considering different objective preferences.

each agent tends to choose actions with higher Q values. Eventually, the reward fluctuation decreases and stabilizes at a higher value. In the three cases, the QoS remains at a stable position close to 1.0, and the carbon intensity reward also converges to a lower value in the overall fluctuation. The specific value of the carbon intensity reward depends on the job duration of the day and the number of GPUs required. In addition, through horizontal comparison, we can find that the number of jobs and the number of iterations required for convergence are positively correlated. Case 1 converges the fastest, with a trend of convergence of around 400 iterations, and then stabilizes at around 700 iterations after a certain fluctuation. Case 2 has greater volatility than Case 1 and converges at around 800 iterations. Case 3 has a slower convergence speed because the number of jobs on that day reaches thousands, requiring about 1,800 iterations.

Overall, for the two reward targets, due to different evolutionary directions, the policy gradients during training sometimes overlap and sometimes diverge. These targets often show a trade-off relationship during training, and they will affect each other and cannot reach the optimal state at the same time, but there are also cases where they move forward together along the same policy gradient direction, especially in the iteration stage before convergence. The two reward functions show a clear convergence trend in the three cases, indicating that the optimization process effectively achieves a stable state of carbon intensity and QoS rewards. We apply a consistent reward structure and agent configuration to all cases without case-specific tuning, and training typically converges within 10 to 70 minutes, depending on workload scale (e.g., over 1,000 jobs in Case 3).

C. Analysis of the Effectiveness of the Model

To further evaluate the effectiveness of the scheduling model, we compare the decision-making behaviors under different objective preferences derived from the Pareto front (Fig. 6). Specifically, we select two representative solutions from distinct regions—one prioritizing QoS and the other emphasizing carbon emission reduction—and compare them to the real-world data center scheduling logs. We compare the scheduling results of June 7 (Case 1) with the scheduling log scheme of the same day, which from the operational data of the data center, which did not consider carbon emission impact at all, is used as the base case benchmark, as shown in Fig. 5. Due to the trade-off characteristics of multi-objective optimization, different schedule results can be obtained by adjusting the weighting between maximizing QoS and minimizing carbon emissions. Thus, to illustrate this, two different scheduling schemes are selected here. The scheduling scheme on the left ((a) and (c)) focuses more on improving QoS, that is, trying to schedule the arriving jobs as soon as possible, while the scheduling scheme on the right ((b) and (d)) focuses more on carbon emissions, that is, trying to avoid executing jobs when the carbon intensity curve is high.

Considering that the jobs that arrive later on the same day will continue to run on the next day, we observed the scheduling of the same day and the operation of the data center in the first 12 hours of the next day, i.e., 36 hours. The figure

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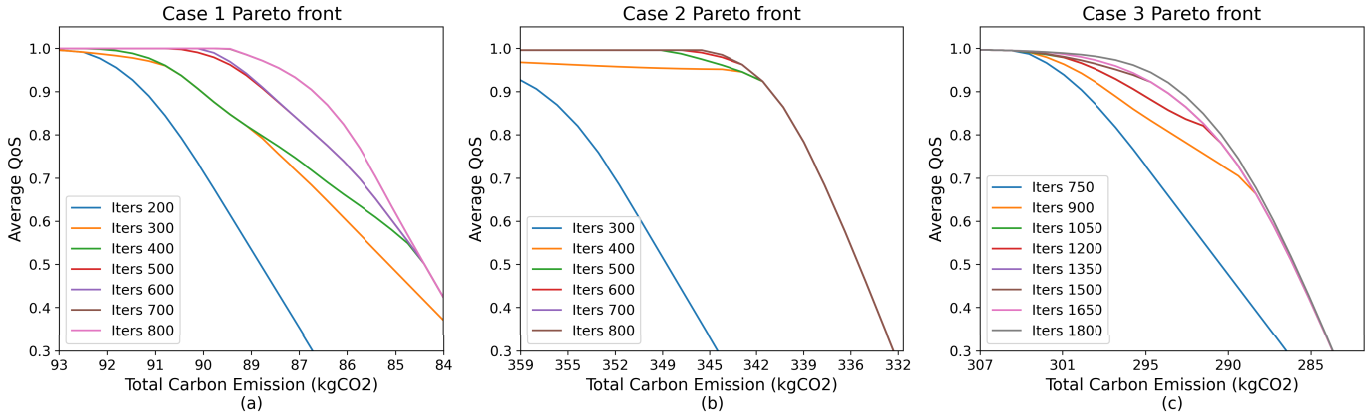


Fig. 6. Data center scheduling Pareto front results.

above ((a) and (b)) compares the number of GPUs running in the two time periods for the scheduling scheme provided by our proposed model and the scheduling log scheme, where the blue is the scheduling log scheme, and the orange is the scheduling scheme obtained using our proposed method.

To show the difference more intuitively, we calculated the difference between the two curves and displayed them as a bar chart on the lower side. The olive green line is the carbon intensity curve.

By comparing the bottom left and right images ((c) and (d)), the left image (c) focuses more on QoS. In this case, jobs are still prioritized even during the higher electricity carbon intensity in the afternoon and evening. On the right side (d), which emphasizes lower carbon emissions, job scheduling is adjusted to avoid peak.

The GPU ID in the log file is hashed, and the model is not provided. We refer to the execution environment information for the trace analysis platform and GPU compute node in the data center [25] and assume that the model is NVIDIA TU102 [GeForce RTX 2080 Ti Rev. A] with a power of 600W. In this case, the scheduling log scheme generates approximately 115.46 kg CO2, while our two scheduling schemes have carbon emissions of about 87.55 and 86.46 kg, resulting in a reduction of about 24%. The average QoS of the jobs in the scheduling log scheme is around 0.90, whereas in our example, they are 0.93 and 0.82, respectively.

D. Analysis of Iteration Result

In the previous section, we demonstrated the effectiveness of the model by showcasing two schedule results with different preferences between the two optimization objectives. Moreover, the method proposed can actually generate a range of alternative schemes that effectively construct the Pareto frontier. Here, we numerically analyzed the Pareto frontier for different reward weights on three selected days.

Fig. 6 shows the Pareto frontier, with each subgraph representing a different case. In each subgraph, the horizontal axis represents the carbon intensity reward, and the vertical axis represents the QoS reward, whose value is the average of all jobs. Therefore, based on the definition of QoS, the maximum value of the vertical axis is 1, and the order of magnitude of the horizontal axis depends on the case's situation on case day. Each curve in the figure represents the results obtained at different iterations.

The left figure illustrates the Pareto frontier of Case 1, with the number of iterations ranging from 200 to 800. In this case, with fewer jobs on that day, achieving a higher QoS is relatively easier. As the number of iterations increases, the Pareto frontier shifts to the right to optimize the target of reducing carbon emissions.

The middle figure displays the Pareto frontier of Case 2, with the number of iterations ranging from 300 to 800. In contrast to Case 1, this case involves more jobs arriving on the day, requiring more training to achieve a high QoS.

The right figure shows the Pareto frontier of Case 3. With thousands of jobs to be completed, the number of iterations required for training is higher, ranging from 750 to 1800. Unlike the first two cases, due to the large number of jobs, the average QoS of the jobs can no longer reach 1.

The diversity of the generated Pareto frontier demonstrates that the ensemble of base agents, each trained with a different fixed λ preference, effectively covers a wide range of trade-offs between carbon emission reduction and QoS objectives.

The combined comparison shows that all three cases show a tendency to converge as the number of iterations continues to rise. In the case of the heavier jobs, the number of convergences to be achieved is simultaneously higher, which is consistent with the conclusion in Section IV-B. The range of the converged Pareto frontier empowers decision-makers to flexibly balance QoS and carbon emissions based on actual needs.

V. Conclusion

This paper proposes a multi-objective scheduling strategy for data centers, with the goal of minimizing carbon emissions and maximizing QoS by finding the Pareto frontier. The proposed solution is an integrated reinforcement learning model based on D3QN. The method uses multiple D3QN agents to dynamically integrate action strategies of different objectives through various weight combinations. These agents explore scheduling schemes in different dimensions of the entire action

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space and gradually approach the Pareto frontier of the multi-objective scheduling scheme.

The simulation of data center scheduling considers multiple factors such as virtual cluster, QoS, and carbon intensity of electricity supply to obtain the Pareto frontier. To verify the effectiveness of the model, real data center logs of SenseTime Technology are used as experimental data. The results show that the proposed method can adaptively learn dual-objective job scheduling strategies, show good convergence results, and finally obtain the Pareto frontier for decision-makers. For example, on June 7, the proposed solution achieved a reduction of 24% in carbon emissions in data centers while ensuring QoS.

The proposed solution provides a practical approach for the low-carbon operation of data centers suitable for real-world online deployment. It provides multiple solutions, additional decision support, and promotion of coordination between data centers and power systems.

Future work could explore enhancing the proactive interaction between data center operations and the electricity grid through the active provision of demand response services. These services would be accessible not only during periods of low-carbon electricity but also when the grid necessitates ancillary services such as frequency response. The proposed multi-objective online scheduling framework is also applicable in advancing this research direction.

REFERENCES

[1] M. H. B. Mahbod, C. B. Chng, P. S. Lee, and C. K. Chui, "Energy saving evaluation of an energy efficient data center using a model-free reinforcement learning approach," *Appl. Energy*, vol. 322, Sep. 2022, Art. no. 119392, doi: 10.1016/j.apenergy.2022.119392.

[2] I. E. Agency. (2023). *Data Centres and Data Transmission Networks*. [Online]. Available: https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks

[3] E. Xing and A. Monck. (2023). *Ai Emissions Are Fueling a New Doomerism. This Time It's Climate Change*. [Online]. Available: https://fortune.com/2023/12/12/ai-emissions-doomerism-climate-change-environment-tech

[4] U.S. Government Accountability Office. (Feb. 2023). *Data Center Optimization: Agencies Continue to Report Progress*. [Online]. Available: https://www.gao.gov/products/gao-23-105946

[5] E. Masanet, A. Shehabi, N. Lei, S. Smith, and J. Koomey, "Recalibrating global data center energy-use estimates," *Science*, vol. 367, no. 6481, pp. 984–986, Feb. 2020, doi: 10.1126/science.aba3758.

[6] M. Uddin and A. A. Rahman, "Energy efficiency and low carbon enabler green IT framework for data centers considering green metrics," *Renew. Sustain. Energy Rev.*, vol. 16, no. 6, pp. 4078–4094, Aug. 2012, doi: 10.1016/j.rser.2012.03.014.

[7] A. Radovanovic et al., "Carbon-aware computing for datacenters," *IEEE Trans. Power Syst.*, vol. 38, no. 2, pp. 1270–1280, Mar. 2023, doi: 10.1109/TPWRS.2022.3173250.

[8] A. Khosravi, L. L. H. Andrew, and R. Buyya, "Dynamic VM placement method for minimizing energy and carbon cost in geographically distributed cloud data centers," *IEEE Trans. Sustain. Comput.*, vol. 2, no. 2, pp. 183–196, Apr. 2017, doi: 10.1109/TSUSC.2017.2709980.

[9] R. Gupta, S. Asgari, H. Moazamigoodarzi, D. G. Down, and I. K. Puri, "Energy, exergy and computing efficiency based data center workload and cooling management," *Appl. Energy*, vol. 299, Oct. 2021, Art. no. 117050, doi: 10.1016/j.apenergy.2021.117050.

[10] L. Yu, T. Jiang, Y. Cao, and Q. Qi, "Joint workload and battery scheduling with heterogeneous service delay guaranteesfor data center energy cost minimization," *IEEE Trans. Parallel Distrib. Syst.*, vol. 26, no. 7, pp. 1937–1947, Jul. 2015, doi: 10.1109/TPDS.2014.2329491.

[11] V. K. Vavilapalli et al., "Apache Hadoop YARN: Yet another resource negotiator," in *Proc. 4th Annu. Symp. Cloud Comput.*, 2013, pp. 1–16, doi: 10.1145/2523616.2523633.

[12] B. Tripathy, S. Dash, and S. K. Padhy, "Dynamic task scheduling using a directed neural network," *J. Parallel Distrib. Comput.*, vol. 75, pp. 101–106, Jan. 2015, doi: 10.1016/j.jpdc.2014.09.015.

[13] D. Fernández-Cerero, J. A. Troyano, A. Jakóbik, and A. Fernández-Montes, "Machine learning regression to boost scheduling performance in hyper-scale cloud-computing data centres," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 6, pp. 3191–3203, Jun. 2022, doi: 10.1016/j.jksuci.2022.04.008.

[14] M. Sharma and R. Garg, "An artificial neural network based approach for energy efficient task scheduling in cloud data centers," *Sustain. Comput., Informat. Syst.*, vol. 26, Jun. 2020, Art. no. 100373, doi: 10.1016/j.suscom.2020.100373.

[15] S. Liang, Z. Yang, F. Jin, and Y. Chen, "Data centers job scheduling with deep reinforcement learning," in *Proc. Pacific–Asia Conf. Knowl. Discovery Data Mining*, 2020, pp. 906–917, doi: 10.1007/978-3-030-47436-2_68.

[16] D. Ding, X. Fan, Y. Zhao, K. Kang, Q. Yin, and J. Zeng, "Q-learning based dynamic task scheduling for energy-efficient cloud computing," *Future Gener. Comput. Syst.*, vol. 108, pp. 361–371, Jul. 2020, doi: 10.1016/j.future.2020.02.018.

[17] W. Liu et al., "Online job scheduling scheme for low-carbon data center operation: An information and energy Nexus perspective," *Appl. Energy*, vol. 338, May 2023, Art. no. 120918, doi: 10.1016/j.apenergy.2023.120918.

[18] X. Zhao and C. Wu, "Large-scale machine learning cluster scheduling via multi-agent graph reinforcement learning," *IEEE Trans. Netw. Service Manage.*, vol. 19, no. 4, pp. 4962–4974, Dec. 2022, doi: 10.1109/TNSM.2021.3139607.

[19] S. Zhang, M. Xu, W. Y. Bryan Lim, and D. Niyato, "Sustainable AIGC workload scheduling of geo-distributed data centers: A multi-agent reinforcement learning approach," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, Dec. 2023, pp. 3500–3505, doi: 10.1109/globecom54140.2023.10437617.

[20] I. Riepin, T. Brown, and V. M. Zavala, "Spatio-temporal load shifting for truly clean computing," *Adv. Appl. Energy*, vol. 17, Mar. 2025, Art. no. 100202, doi: 10.1016/j.adapen.2024.100202.

[21] Y. Lian, Y. Li, Y. Zhao, C. Yu, T. Zhao, and L. Wu, "Robust multi-objective optimization for islanded data center microgrid operations," *Appl. Energy*, vol. 330, Jan. 2023, Art. no. 120344, doi: 10.1016/j.apenergy.2022.120344.

[22] D. Yi, X. Zhou, Y. Wen, and R. Tan, "Efficient compute-intensive job allocation in data centers via deep reinforcement learning," *IEEE Trans. Parallel Distrib. Syst.*, vol. 31, no. 6, pp. 1474–1485, Jun. 2020, doi: 10.1109/TPDS.2020.2968427.

[23] J. Yan et al., "Energy-aware systems for real-time job scheduling in cloud data centers: A deep reinforcement learning approach," *Comput. Electr. Eng.*, vol. 99, Apr. 2022, Art. no. 107688, doi: 10.1016/j.compeleceng.2022.107688.

[24] B. Li, S. Samsi, V. Gadepally, and D. Tiwari, "Clover: Toward sustainable AI with carbon-aware machine learning inference service," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, Nov. 2023, pp. 1–15, doi: 10.1145/3581784.3607034.

[25] Q. Hu, P. Sun, S. Yan, Y. Wen, and T. Zhang, "Characterization and prediction of deep learning workloads in large-scale GPU datacenters," in *Proc. Int. Conf. High Perform. Comput., Netw., Storage Anal.*, Nov. 2021, pp. 1–15, doi: 10.1145/3458817.3476223.

[26] P. M. Pardalos, A. Migdalas, and L. Pitsoulis, *Pareto Optimality, Game Theory and Equilibria*, vol. 17. Cham, Switzerland: Springer, 2008, doi: 10.1007/978-0-387-77247-9.

[27] M. Hessel et al., "Rainbow: Combining improvements in deep reinforcement learning," in *Proc. AAAI Conf. Artif. Intell.*, 2017.

[28] H. v. Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *Proc. AAAI Conf. Artif. Intell.*, 2016, vol. 30, no. 1, pp. 2094–2100.

[29] Z. Wang, T. Schaul, M. Hessel, H. v. Hasselt, M. Lanctot, and N. D. Freitas, "Dueling network architectures for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1995–2003.

[30] A. Verma, L. Pedrosa, M. Korupolu, D. Oppenheimer, E. Tune, and J. Wilkes, "Large-scale cluster management at Google with Borg," in *Proc. 10th Eur. Conf. Comput. Syst.*, Apr. 2015, pp. 1–17, doi: 10.1145/2741948.2741964.

[31] ESO. (2020). *Historic Generation Mix and Carbon Intensity*. National Grid ESO. [Online]. Available: https://www.nationalgrideso.com/data-portal/historic-generation-mix

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Yifan Wang (Graduate Student Member, IEEE) received the B.Sc. degree in mathematics from the University of Liverpool, U.K., in 2021, and the M.Sc. degree in scientific and data intensive computing from University College London, U.K., in 2022. He is currently pursuing the Ph.D. degree with the School of Engineering, The University of Edinburgh, U.K. His research interests include the application of machine learning techniques in energy systems, with a focus on optimisation, system operation, and energy sustainability.



Wei Sun (Member, IEEE) received the Ph.D. degree in power system engineering from The University of Edinburgh, U.K. He is currently a Chancellor's Fellow on energy systems integration at The University of Edinburgh. He was a Lead Researcher at the National Centre for Energy Systems Integration and has contributed to several EPSRC-funded projects related to energy storage, energy system resilience, and green hydrogen. He is a Chartered Engineer. His research focuses on the development and application of novel optimization methods for the planning and operation of smart energy systems.



Pengyu Ren (Graduate Student Member, IEEE) received the B.S. degree in electrical engineering and automation from Southeast University, Nanjing, China, in 2021. He is currently pursuing the Ph.D. degree in electrical engineering with the School of Engineering, The University of Edinburgh, Edinburgh, U.K. His research interests include power system optimization and stability analysis.



Gareth Harrison (Senior Member, IEEE) is the Bert Whittington Chair of electrical power engineering at The University of Edinburgh. He has held a number of executive leadership roles in the School of Engineering, most recently as the Head of the School of Engineering. He is currently a co-investigator on several EPSRC and Ofgem/InnovateUK projects covering energy storage and power networks. He was the Associate Director of the EPSRC National Centre for Energy Systems Integration (2016–2022), a Principal Investigator of the EPSRC Adaptation and Resilience in Energy Systems Project (2012–2015), examining climate impacts on energy system resilience, and has been a co-investigator on many U.K. and EU consortia. He is a Chartered Engineer. His research is focused on renewable energy integration within energy systems, and he has built a reputation for the application of meteorological modeling and optimisation methods for energy network operation and planning. He is a fellow of the Institution of Engineering and Technology.

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